QUES1- WRITE A PROGRAM TO DO ARITHMETIC OPERATIONS ON TWO GIVEN NUMBERS.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
sum=a+b
print("Sum is =", sum)
```

```
Enter first number: 12
Enter second number: 13
Sum is = 25
```

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
difference=a-b
print("difference is =", difference)
```

Output

```
Enter first number: 12
Enter second number: 13
difference is = -1
```

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
division=a/b
print(division)
```

Output

```
Enter first number: 30
Enter second number: 12
2.5
```

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
remainder=a%b
print(remainder)
```

```
Enter first number: 20
Enter second number: 10
0
```

QUES2- FIND THE SIMPLE INTEREST.

```
P = int(input("Enter principal: "))
R = int(input("Enter rate: "))
T = int(input("Enter time: "))
SI=P*R*T/100
print(SI)
```

```
Enter principal: 20000
Enter rate: 12
Enter time: 4
9600.0
```

QUES3-FIND THE SUM AND REVERSE OF 3 DIGIT NUMBER.

```
n= int(input("enter a three digit number: "))
d1=n%10
n=n//10
d2=n%10
n=n//10
d3=n%10
n=n//10
sum=d1+d2+d3
reverse=100*d1+10*d2+d3
print(sum)
print(reverse)
```

Output

```
enter a three digit number: 254
11
452
```

QUES4- FIND THE AREA OF THE TRIANGLE?

```python
base = float(input("Enter the base: "))
height = float(input("Enter the height: "))
area = 0.5 * base * height
print("Area of triangle =", area)
```

Output

```
Enter the base: 12
Enter the height: 4
Area of triangle = 24.0
```

QUES5- Write a Python program to calculate the total electricity bill based on the number of units consumed, using the following rates:

- If units are less than 100, the rate is ₹3 per unit.
- If units are between 100 and 300, the rate is ₹4 per unit.
- If units are more than 300, the rate is ₹5 per unit.

```
a = int(input("Enter the amount of unit used: "))
if a <= 100:
    print("The price to be paid per unit is 3 rupees")
    print("Total amount to be paid: ", a * 3)
elif a <= 300 and a > 100:
    print("The price to be paid per unit is 4 rupees")
    print("Total amount to be paid: ", a * 4)
else:
    print("The price to be paid per unit is 5 rupees")
    print("Total amount to be paid: ", a * 5)
```

Output

```
Enter the amount of unit used: 150
The price to be paid per unit is 4 rupees
Total amount to be paid:  600
```

QUES6- TO CHECK WHETHER THE NUMBER IS ODD OR EVEN.

```
a = int(input("Enter a number: "))
if a % 2 == 0:
    print("The number is even")
else:
    print("The number is odd")
```

```
Output

Enter a number: 13
The number is odd
```

QUES7 - TO FIND SMALLEST AND GREATEST AMONG THREE-DIGITS.

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a > b and a > c:
    greatest = a
elif b > a and b > c:
    greatest = b
else:
    greatest = c
if a < b and a < c:
    smallest = a
elif b < a and b < c:
    smallest = b
else:
    smallest = c
print("Greatest number =", greatest)
print("Smallest number =", smallest)
```

```
Output

Enter first number: 12
Enter second number: 58
Enter third number: 98
Greatest number = 98
Smallest number = 12
```

QUES8 - TO FIND THE SUM OF N NATURAL NUMBERS.

```python
n = int(input("Enter a number: "))
sum = 0
for i in range(1, n+1):
    sum = sum + i
print("Sum of natural numbers =", sum)
```

```
Output

Enter a number: 12
Sum of natural numbers = 78
```

QUES9–TO FIND FACTORIAL OF N.

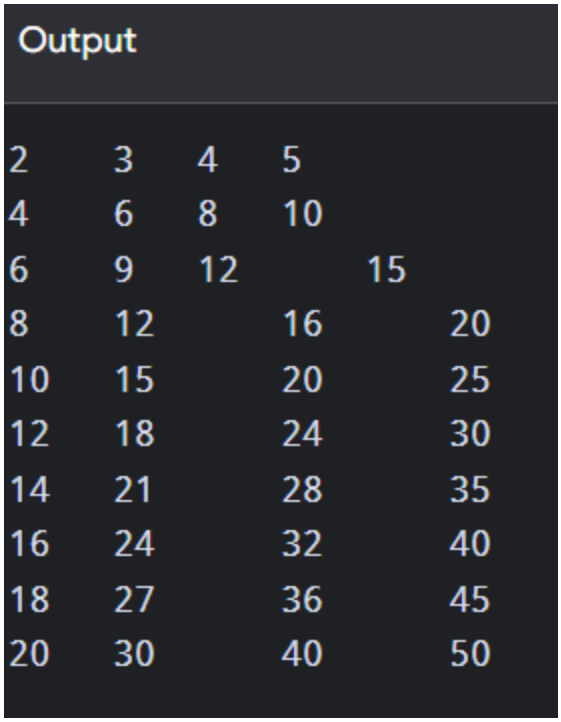
```
n=int(input("enter the number: "))
fact=1
for i in range(1,n):
  fact=fact*i
print(fact)
```

Output

```
enter the number: 6
120
```

QUES10 - TO PRINT MULTIPLICATION TABLE.

```
for i in range(1, 11):
    print(2*i, "\t", 3*i, "\t", 4*i,"\t", 5*i)
```



QUES11 - OPERATIONS ON STACK:
CREATE
PUSH

POP

DISPLAY

UNDERFLOW

OVERFLOW

```python
S = []
n = int(input("Enter the number: "))

for i in range(n):
    x = int(input("Enter the number to be inserted in stack: "))
    S.append(x)
print("The Stack is:", S)

if len(S) == 0:
    print("The stack is empty")
else:
    print("The stack is not empty")

a = input("Do you want to add element (Yes/No): ")
if a == "Yes":
    e = int(input("Enter number of elements you want to add: "))
    for i in range(e):
        x2 = int(input("Enter element: "))
        S.append(x2)
    print("The new stack is:", S)

b = input("Do you want to delete element (Yes/No): ")
if b == "Yes":
    if len(S) == 0:
        print("Stack Underflow")
    else:
        S.pop()
        print("The new stack is:", S)

print("The final stack is:", S)
```

```
Enter the number: 4
Enter the number to be inserted in stack: 1
Enter the number to be inserted in stack: 2
Enter the number to be inserted in stack: 3
Enter the number to be inserted in stack: 4
The Stack is: [1, 2, 3, 4]
The stack is not empty
Do you want to add element (Yes/No): Yes
Enter number of elements you want to add: 2
Enter element: 8
Enter element: 9
The new stack is: [1, 2, 3, 4, 8, 9]
Do you want to delete element (Yes/No): Yes
The new stack is: [1, 2, 3, 4, 8]
The final stack is: [1, 2, 3, 4, 8]
```

QUES12 - TO FIND REVERSE OF A STACK

```
a=[]
n=int(input("enter the number: "))
for i in range(n):
  x=int(input("enter the no: "))
  a.append(x)
print("The given stack is:",a)

rev=[]
while len(a)>0:
  x=a.pop()
  rev.append(x)
print("the reverse of the stack is: ", rev)
```

```
enter the number: 5
enter the no: 1
enter the no: 2
enter the no: 3
enter the no: 4
enter the no: 5
The given stack is: [1, 2, 3, 4, 5]
the reverse of the stack is:  [5, 4, 3, 2, 1]
```

QUES13 - TO MATCH PARENTHESES

```
S = []
a = input("Enter an expression: ")
for i in a:
    if i == '(':
        S.append(i)

    elif i == ')':
      if len(S) == 0:
        S.append("Error")
        break
      x = S.pop()

      if (i == ')' and x != '('):
        S.append("Error")
        break
if len(S) == 0:
    print("Parentheses are balanced")
else:
    print("Parentheses are not balanced")
```

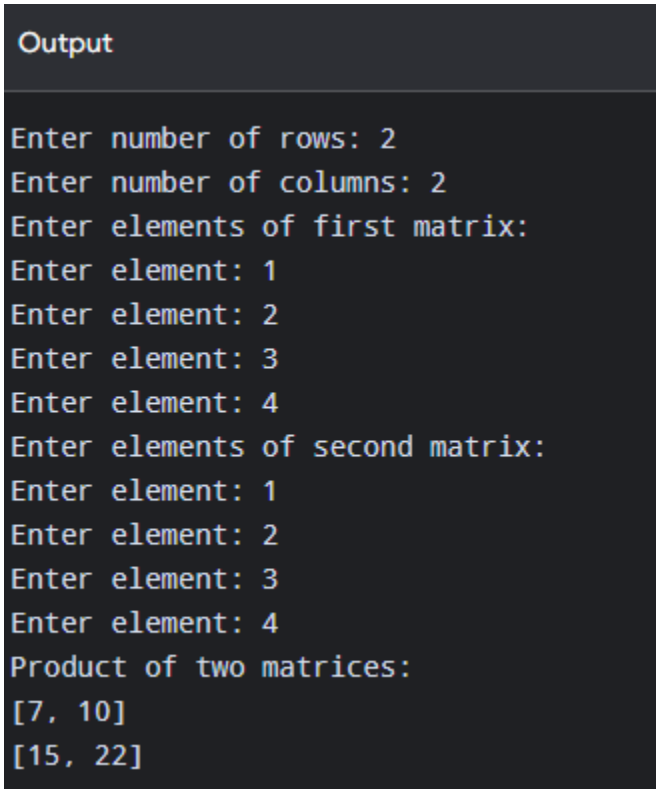
Output

```
Enter an expression: (A+B(C
Parentheses are not balanced
```

QUES14- TO FIND PRODUCT OF TWO MATRICES

```
A = []
B = []
r = int(input("Enter number of rows: "))
c = int(input("Enter number of columns: "))
print("Enter elements of first matrix:")
for i in range(r):
    row = []
    for j in range(c):
        x = int(input("Enter element: "))
        row.append(x)
    A.append(row)

print("Enter elements of second matrix:")
for i in range(r):
    row = []
    for j in range(c):
        x = int(input("Enter element: "))
        row.append(x)
    B.append(row)
C = []
for i in range(r):
    row = []
    for j in range(c):
        s = 0
        for k in range(c):
            s = s + A[i][k] * B[k][j]
        row.append(s)
    C.append(row)
print("Product of two matrices:")
for i in range(r):
    print(C[i])
```



QUES15 - TO FIND THE SUM OF DIAGONAL ELEMENTS.

```python
A = []

n = int(input("Enter the order of matrix: "))

print("Enter matrix elements:")

for i in range(n):
    row = []
    for j in range(n):
        x = int(input("Enter element: "))
        row.append(x)
    A.append(row)

diagonal = 0
non_diagonal = 0

for i in range(n):
    for j in range(n):
        if i == j:
            diagonal = diagonal + A[i][j]
        else:
            non_diagonal = non_diagonal + A[i][j]

print("Sum of diagonal elements =", diagonal)
print("Sum of non-diagonal elements =", non_diagonal)
```

Output

```
Enter the order of matrix: 2
Enter matrix elements:
Enter element: 1
Enter element: 2
Enter element: 3
Enter element: 4
Sum of diagonal elements = 5
Sum of non-diagonal elements = 5
```

QUES16 - Enter marks of 5 students for 3 subjects each and find the percentage of students.

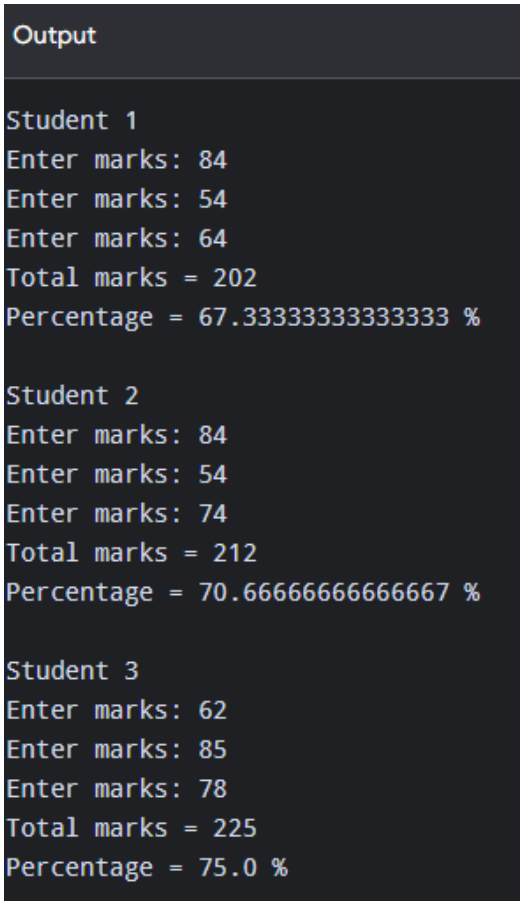
```
for i in range(1, 4):
    print("Student", i)

    total = 0

    for j in range(1, 4):
        marks = int(input("Enter marks: "))
        total = total + marks

    percentage = total / 300 * 100

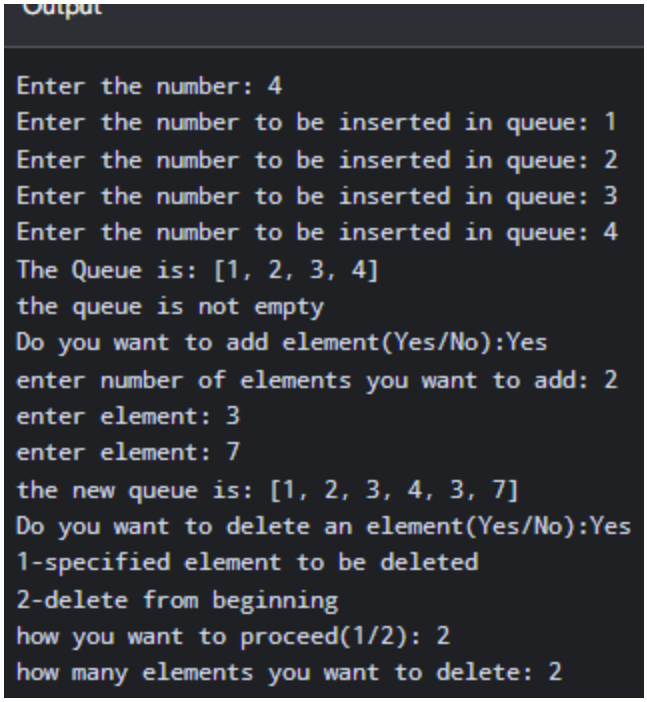
    print("Total marks =", total)
    print("Percentage =", percentage, "%")
    print()
```



QUES17 - TO PERFORM OPERATIONS OF QUEUE.

```python
Q = []
n = int(input("Enter the number: "))
for i in range(n):
    x = int(input("Enter the number to be inserted in queue: "))
    Q.append(x)
print("The Queue is:", Q)
# Checking whether the queue is empty or not
if len(Q) == 0:
    print("the queue is empty")
else:
    print("the queue is not empty")
# Inserting an element into the queue
a = input("Do you want to add element(Yes/No):")
if a == "Yes":
    e = int(input("enter number of elements you want to add: "))
    for i in range(e):
        x2 = int(input("enter element: "))
        Q.append(x2)
else:
    pass
print("the new queue is:", Q)
# Deleting an element
b = input("Do you want to delete an element(Yes/No):")
if b == "Yes":
    print("1-specified element to be deleted")
    print("2-delete from beginning")
    ch = int(input("how you want to proceed(1/2): "))
    if ch == 1:
        a = int(input("enter the element you want to delete: "))
        Q.pop(a)
        print("After deletion queue:", Q)
    if ch == 2:
        e2 = int(input("how many elements you want to delete: "))
        while len(Q) > 0:
            for i in range(e2):
                Q.pop(i)
else:
    print("no element to be deleted")

# Displaying the final queue
print("The final queue is:", Q)
```



QUES18 - CIRCULAR QUEUE

```python
Q = []
n = int(input("Enter the size of circular queue: "))
front = -1
rear = -1
for i in range(n):
    x = int(input("Enter element to insert: "))
    Q.append(x)
front = 0
rear = n - 1
print("The Circular Queue is:", Q)
if len(Q) == 0:
    print("The queue is empty")
else:
    print("The queue is not empty")
a = input("Do you want to add element (Yes/No): ")

if a == "Yes":
    if (rear + 1) % n == front:
        print("Circular Queue Overflow")
    else:
        x = int(input("Enter element: "))
        rear = (rear + 1) % n
        Q[rear] = x
        print("The new circular queue is:", Q)
b = input("Do you want to delete an element (Yes/No): ")
if b == "Yes":
    if front == -1:
        print("Circular Queue Underflow")
    else:
        print("Deleted element:", Q[front])
        Q[front] = None

        if front == rear:
            front = -1
            rear = -1
        else:
            front = (front + 1) % n
        print("Queue after deletion:", Q)
print("The final circular queue is:", Q)
```

Output

```
Enter the size of circular queue: 4
Enter element to insert: 2
Enter element to insert: 5
Enter element to insert: 7
Enter element to insert: 9
The Circular Queue is: [2, 5, 7, 9]
The queue is not empty
Do you want to add element (Yes/No): Yes
Circular Queue Overflow
Do you want to delete an element (Yes/No): Yes
Deleted element: 2
Queue after deletion: [None, 5, 7, 9]
The final circular queue is: [None, 5, 7, 9]
```